

ROUSHAN KUMAR

Systems Engineer | Backend Architect | AI Developer Tools Specialist

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PROFESSIONAL SUMMARY

Systems-focused Software Engineer with deep expertise in building low-latency REST APIs, concurrent microservice architectures, and full-stack web applications. Proficient across C#/.NET Core, Java Spring Boot, Python FastAPI, and modern TypeScript runtimes (Node.js, Express, React, Next.js), with a proven track record of architecting zero-knowledge cryptographic vaults and autonomous AI developer sandboxes. Demonstrated mastery in database normalization (3NF), asynchronous event loops, and automated test automation (xUnit, JUnit 5, PyTest) to deliver robust, enterprise-grade software products.

TECHNICAL SKILLS

Backend & Systems: C# (.NET Core Web API) • Java (Spring Boot) • Python (FastAPI) • Node.js • Fastify • Express.js • REST APIs

Databases & Caching: PostgreSQL • MongoDB • Redis • SQLite • Supabase • Schema Design (3NF)

Languages: C • Rust • C# • Java • Python • JavaScript • TypeScript • SQL • HTML5 • CSS3

Frontend UI & Full Stack: React.js • Next.js • Express.js • Redux • Monaco Editor • TailwindCSS • Bootstrap

Testing & DevOps: xUnit • Moq • JUnit 5 • PyTest • Jest • Docker • CI/CD • GCP • Git • Gradle • Vercel • Render

AI Tooling & Security: Gemini API • Tree-Sitter AST • Web Crypto API (AES-GCM) • PBKDF2 • JWT • Prompt Engineering

TECHNICAL PROJECTS

Astra Vision - Developer Sandbox & Code Graph Parser (FastAPI, Python, Monaco, Tree-Sitter)

[Live Demo](#) | [GitHub](#)

- **Product Architecture:** Architected an automated code-parsing platform and browser sandbox that allows developers to safely execute untrusted code while visualizing full-stack repository dependency graphs.
- **Sandbox Runtime Challenge:** Built a self-healing Python execution engine with subprocess isolation and security checkpoints to intercept unauthorized OS file system calls and network socket requests in real time.
- **Indexing Performance:** Integrated Tree-Sitter AST compilers and ChromaDB vector search to parse repository syntax nodes, enabling instant code graph dependency analysis under 200ms.

IDBI FinSync - AI-Powered Wealth & Financial Management Engine (Next.js, React, Fastify, Gemini API)

[Live Demo](#) | [GitHub](#)

- **Product Architecture:** Co-created an intelligent personal financial management web app in a monorepo that unifies bank ledgers, investment portfolios, and expense streams into a live interactive dashboard.
- **AI Integration Challenge:** Embedded "Mitra," an interactive AI wealth consultant leveraging Gemini API to analyze spending habits, detect budget anomalies, and deliver personalized financial guidance.
- **Database & Concurrency:** Designed PostgreSQL transaction ledgers validated by Zod schemas and Fastify microservice endpoints, handling concurrent balance updates with zero data loss.

Alyra Lock - Secure Zero-Knowledge Password Vault (React, TypeScript, Express, MongoDB)

[Live Demo](#) | [GitHub](#)

- **Product Architecture:** Engineered a client-side zero-knowledge password vault ensuring master encryption keys remain entirely isolated inside user browser memory, mitigating cloud database leak risks.
- **Cryptographic Engineering:** Implemented Web Crypto API primitives utilizing AES-GCM (256-bit) payload encryption and PBKDF2 key derivation over 100,000 hashing iterations for key security.
- **Performance & Security:** Developed high-throughput Express.js REST API sync endpoints with JWT authentication and strict CORS headers, keeping vault database sync latency under 200ms.

TECHNICAL SIMULATIONS & VIRTUAL EXPERIENCES

Commonwealth Bank (CommBank) - Software Engineering Virtual Simulation

June 2026

- **Problem & Scope:** Diagnosed and resolved silent data overwrite bugs across high-traffic C#/.NET Core financial API controllers handling MongoDB document updates across financial services endpoints.
- **Action Taken:** Extended C# Web API controllers with `$set` atomic operators for partial payload updates and modernized an interactive React/Redux Goal Manager UI component for banking clients.
- **Engineering Outcome:** Authored comprehensive automated test suites using xUnit and Moq, covering complex boundary conditions to guarantee high reliability and transactional consistency.

Y Combinator (YC Startup - Shiptivity) - Software Engineering Virtual Simulation

June 2026

- **Problem & Scope:** Fixed key constraint collisions and slow re-rendering performance on a real-time drag-and-drop Kanban workflow board used by engineering teams to track task state transitions.
- **Action Taken:** Architected dynamic SQLite priority reordering logic using atomic transactions to update priority ranks sequentially and integrated Dragula drag-and-drop state hooks in React.
- **Engineering Outcome:** Reduced UI component re-render cycles by 30% and patched OpenSSL compilation bottlenecks in legacy Node.js Webpack configurations to optimize runtime builds.

Walmart USA - Advanced Software Engineering Virtual Simulation

May 2026

- **Problem & Scope:** Solved indexing overhead and performance bottlenecks in high-volume retail inventory processing queues handling unnormalized data ingestion from legacy spreadsheets.
- **Action Taken:** Implemented a generic K-ary Max Heap structure in Java using fast bitwise shift operators (`<<`, `>>>`) and built 3NF database schemas with design pattern abstractions.
- **Engineering Outcome:** Accelerated queue indexing calculations by 35% and engineered automated Python ETL pipelines using `csv` and `sqlite3` to clean and consolidate disparate spreadsheets.

EDUCATION

Bachelor of Computer Applications (BCA) - Amity University Noida

Expected July 2028

- **Relevant Coursework:** Data Structures & Algorithms (DSA) • Database Management Systems (DBMS) • Operating Systems (OS) • Computer Networks • Object-Oriented Programming (OOP)